

Try Gemini 1.5 models (<https://console.cloud.google.com/vertex-ai/generative/multimodal/create/text>), the latest multimodal models in Vertex AI, and see what you can build with up to a 2M token context window.

Online prediction logging

For AutoML tabular models, AutoML image models, and custom-trained models, you can enable or disable prediction logs during [model deployment](#) (/vertex-ai/docs/predictions/deploy-model-api) or [endpoint creation](#) (/vertex-ai/docs/predictions/deploy-model-api#create-endpoint). This page explains the different types of prediction logs available, and how to enable or disable these logs.

Types of prediction logs

There are several types of prediction logs that you can use to get information from your prediction nodes:

- **Container logging**, which logs the `stdout` and `stderr` streams from your prediction nodes to [Cloud Logging](#) (/logging/docs/overview). These logs are required for debugging.
 - On the `v1` service endpoint, container logging is enabled by default. You can disable it when you deploy a model. You can also disable or enable logging when you [mutate](#) (/vertex-ai/docs/reference/rest/v1/projects.locations.endpoints/mutateDeployedModel) the deployed model.
 - On the `v1beta1` service endpoint, container logging is disabled by default. You can enable it when you deploy a model. You can also disable or enable logging when you [mutate](#) (/vertex-ai/docs/reference/rest/v1/projects.locations.endpoints/mutateDeployedModel) the deployed model.

★ **Note:** The default logging behavior in Python sends outputs to `stderr`, which will appear at the **ERROR** level in Cloud Logging. If you'd like for container logs to appear at the **INFO** level, configure your container logging to send outputs to `stdout`.

- **Access logging**, which logs information like timestamp and latency for each request to Cloud Logging.

On both the `v1` and `v1beta1` service endpoints, access logging is disabled by default. You can enable access logging when you deploy a model to an endpoint.

★ **Note:** If you have [VPC Service Controls](#) (/vpc-service-controls/docs/overview) enabled for your project, access logging isn't supported.

- **Request-response logging**, which logs a sample of online prediction requests and responses to a BigQuery table.

You can enable request-response logging by creating or patching the prediction endpoint.

You can enable or disable each type of log independently.

Prediction log settings

You can enable or disable online prediction logs when you create an endpoint, deploy a model to the endpoint, or mutate a deployed model.

To update the settings for access logs, you must [undeploy your model](#)

([/vertex-ai/docs/reference/rest/v1/projects.locations.endpoints/undeployModel](#)), and then redeploy the model with your new settings.

You can update the settings for container logs without re-deploying your model.

Online prediction at a high rate of queries per second (QPS) can produce a substantial number of logs, which are subject to [Cloud Logging pricing](#) ([/stackdriver/pricing#logging-costs](#)). To estimate the pricing for your online prediction logs, see [Estimating your bills](#) ([/stackdriver/estimating-bills](#)) for logging. To reduce this cost, you can disable prediction logging.

Enable and disable prediction logs

The following examples highlight where to modify the default log settings:

[Console](#) [gcloud](#) ([#gcloud](#)) [REST](#) ([#rest](#))
([#console](#))

When you deploy a model to an endpoint or create a new endpoint in the Google Cloud console, you can specify which types of prediction logs to enable in the **Logging** step. Select the checkboxes to enable **Access logging** or **Container logging**, or clear the checkboxes to disable these logs.

Use the REST API to update the settings for container logs.

Use the REST API to enable request-response logging. The Google Cloud console and gcloud CLI don't support request-response logging configuration.

To see more context about how to deploy models, read [Deploy a model using the Google Cloud console](#) ([/vertex-ai/docs/predictions/deploy-model-console](#)).

Request-response logging and Model Monitoring v1

Request-response logging and [Model Monitoring v1](#) ([/vertex-ai/docs/model-monitoring/overview](#)) use the same BigQuery table on the backend to log incoming requests. To prevent unexpected changes to this BigQuery table, the following limitations are enforced when using both features at the same time:

- If an endpoint has Model Monitoring enabled, you can't enable request-response logging for the same endpoint.
- If you enable request-response logging and then Model Monitoring on the same endpoint, you won't be able to change the request-response logging configuration.

What's next

- [Estimate pricing](#) ([/stackdriver/estimating-bills](#)) for online prediction logging.
- Deploy a model [using the Google Cloud console](#) ([/vertex-ai/docs/predictions/deploy-model-console](#)) or [using the Vertex AI API](#) ([/vertex-ai/docs/predictions/deploy-model-api](#)).
- Learn [how to create a BigQuery table](#) ([/bigquery/docs/tables](#)).

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